WEEK 7 LAB

How muscle makes force.

2 to 3 hours. Graded. Investigate It, 25 percent of your grade across the term.

This week you run PhysioEx Exercise 2, Skeletal Muscle Physiology, where you stimulate an isolated muscle and record its twitches, summation, tetanus, fatigue, length-tension curve and load-velocity curve. Then you tire out your own hand with a clothespin and compare. Last, one clinical case puts the simulation back into a real body.

Write every answer in your own words

1 min

Do not copy from the book, the notes, a website, a classmate or an AI tool, even if you change a few words. Read the source, look away, and then write what you understood. If you cannot say it without looking, that tells you what to go back and study. Your own words, even rough ones, are what I want to see.

What you need

1 min

- PhysioEx, from the Pearson link in Canvas. Open Exercise 2, Skeletal Muscle Physiology.
- For the grip test at home: a spring clothespin, the kind that pinches, and a timer such as the one on your phone. A partner to keep time helps but is not needed. No clothespin? Squeeze a tennis ball or a rolled-up pair of socks in your fist instead, and say which you used.
- Paper and a pen for your two graphs, or the printed worksheet. You can type everything else on this page.

What goes in each box

1 min

1. **What did you measure?** Name what you changed and what you recorded, with units.
2. **What were your findings?** Give the actual results, with numbers. What went up, what went down, and what stayed the same.
3. **What did you learn?** Explain why the results came out that way, using the physiology from this week. This is the most important box.

Your worksheet is graded once PhysioEx shows each activity complete. The PhysioEx Lab Report that Pearson makes is not this worksheet and earns no points.

Part 1. PhysioEx Exercise 2, Skeletal Muscle Physiology

1 min

Run every activity in Exercise 2, in the order PhysioEx lists them. Pearson sometimes renumbers activities, so go by the topic named below.

Activity 1

1 min

The muscle twitch and the latent period

What did you measure?

What were your findings?

What did you learn?

How long was the latent period you measured, in milliseconds, and what is happening inside the fiber during it?

Activity 2

1 min

The effect of stimulus voltage

What did you measure?

What were your findings?

What did you learn?

Your threshold voltage and your maximal voltage. In one or two sentences, say what happens to the force between them, and why, in terms of how many fibers are responding.

Activity 3

1 min

The effect of stimulus frequency: treppe and wave summation

What did you measure?

What were your findings?

What did you learn?

The stimulus frequency, or the time between two stimuli, at which summation began.

Activity 4

1 min

Tetanus

What did you measure?

What were your findings?

What did you learn?

The stimulus frequency at which tetanus was complete, with no dips left between stimuli. What is happening to calcium inside the fiber at that frequency?

Activity 5

1 min

Fatigue

What did you measure?

What were your findings?

What did you learn?

How long it took the force to fall by the amount you chose, and what that amount was. Record the time, not just the shape of the trace.

Activity 6

1 min

The length-tension relationship

What did you measure?

What were your findings?

What did you learn?

Draw it on paper: Your length-tension curve, by hand: muscle length on the x axis, active force on the y axis. Plot every point you took, enough to see both sides of the curve, and mark the resting length.

At which length was active force highest? Why does force fall at lengths shorter and longer than that?

Activity 7

1 min

Isotonic contractions and the load-velocity relationship

What did you measure?

What were your findings?

What did you learn?

At which load did the muscle stop shortening, and what kind of contraction was it then?

Part 2. Grip fatigue, at home

1 min

Squeeze until your hand tires

2 min

PhysioEx shows fatigue in an isolated muscle. This part shows it in yours. There is no surface EMG at home, so you count instead, and Part 2 asks what an EMG would have shown.

1. Hold the clothespin between the tip of your thumb and the tip of your index finger of the hand you write with. Rest your forearm on a table.
2. Start the timer. Squeeze the clothespin all the way closed and let it open, as fast as you can, for 2 minutes without stopping.
3. Count the squeezes in every 15 seconds, or have your partner count while you squeeze, and write each count in the table.
4. Put the clothespin down and rest for 2 minutes. Then count one more 15 seconds.

A burning feeling in your forearm is expected and goes away with rest. Stop if you feel sharp pain, and skip this part if you have a hand or wrist injury. Say so on your worksheet instead.

TIME	SQUEEZES
0 to 15 seconds	
15 to 30 seconds	
30 to 45 seconds	
45 to 60 seconds	
60 to 75 seconds	
75 to 90 seconds	
90 to 105 seconds	
105 to 120 seconds	

TIME	SQUEEZES
After 2 minutes of rest, 15 seconds	

Draw it on paper: Your fatigue graph, by hand: time on the x axis and squeezes per 15 seconds on the y axis, with the after-rest count as a separate point.

Describe what happened to your count over the 2 minutes, with numbers. By how much did it fall from the first 15 seconds to the last?

After 2 minutes of rest, how much came back? What does that tell you about the cause: something that recovers in minutes, or damage to the muscle?

Suppose you had held the clothespin closed with the same steady force instead of squeezing it over and over. For a while the force would stay the same even as the muscle tired. What would your nervous system have to do to keep the force the same, and what would a surface EMG recording show as it did?

Name one cause of fatigue inside the muscle fibers and one cause in the nervous system. Which one do you think limited your count, and why?

Part 3. Clinical correlation

1 min

Sore on Wednesday, not on Monday

1 min

An athlete does a session of downhill running on Monday and feels fine that evening. Soreness begins Tuesday afternoon, peaks Wednesday and is gone by Friday. Blood lactate returns to its resting level within about an hour of any hard session. Downhill running loads the thigh muscles while they are lengthening.

Use the lactate time course to rule something out. Say plainly what it rules out.

Name what a lengthening contraction does mechanically that a shortening one does not, and connect it to your length–tension curve from Activity 6.

Explain the delay of about two days in terms of what has to happen after the damage before you feel it.

Look back at your thinking

1 min

What did you predict tetanus would look like before you produced one?

Where on your own length–tension curve did you expect peak force, and where was it?

Say one thing about muscle you believed on Monday that you would now state differently.

Turn it in

1 min

Save this page as a PDF with the button below. Photograph your two graphs. If you worked on the printed worksheet, scan or photograph every page instead. Combine everything into one PDF and upload it to the Week 7 lab assignment in Canvas. Due Sunday, October 25 at 10:00 pm.